



Computer Organisation & Program Execution 2019

Charles Martin & Uwe R. Zimmer - The Australian National University

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Organization & Contents

Uwe R. Zimmer - The Australian National University

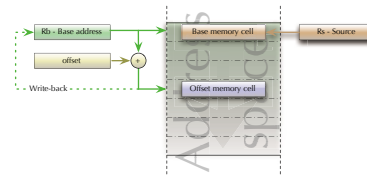
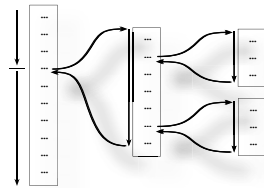


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what is offered here?

```

b    while_condition
while:
mul  r1, r1
while_condition:
cmp  r1, #100
blt  while
    
```

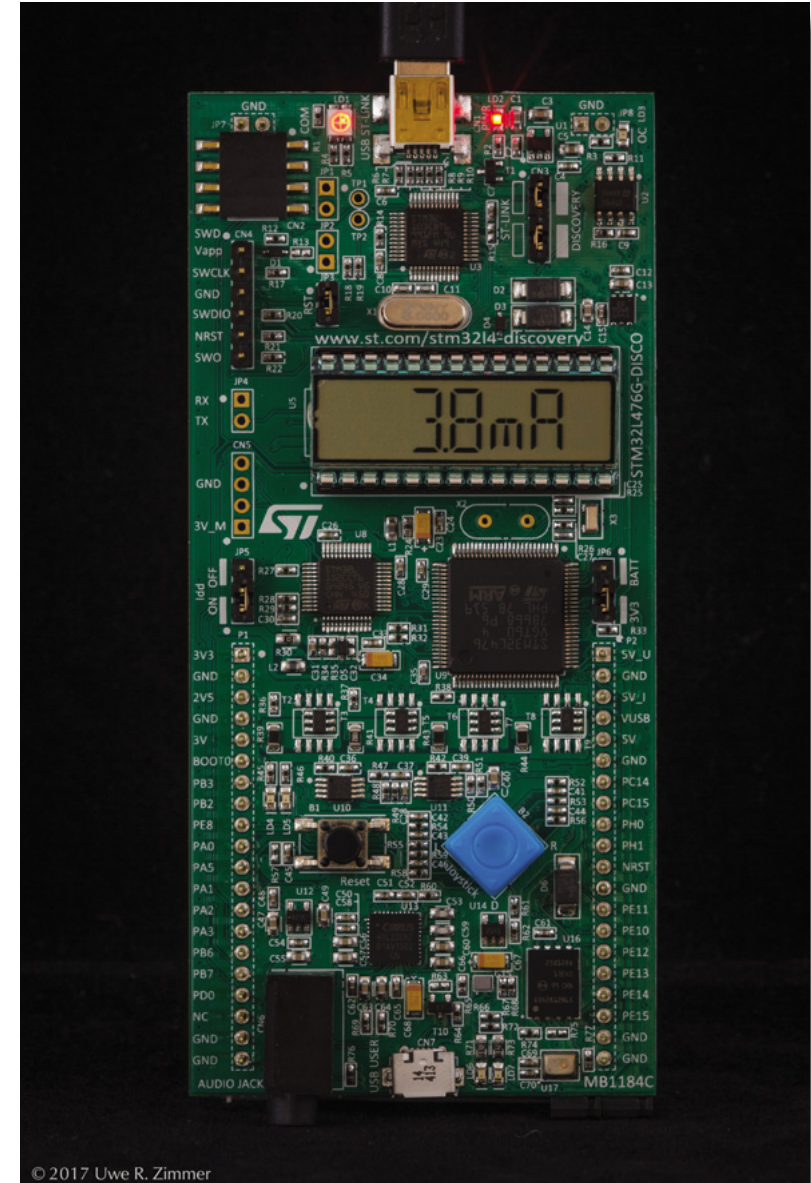
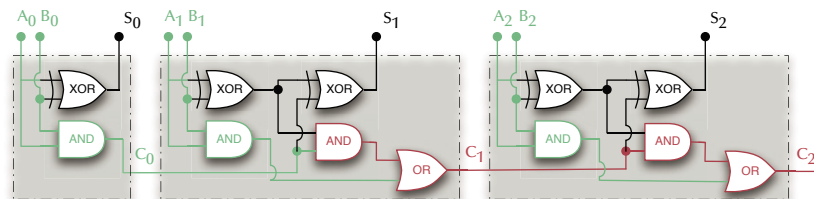


Fundamentals, Overview & Hands-on Experience of Computer Architecture

ADDS <Rd>, <Rn>, <Rm>

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	1	1	0	0			Rm		Rn				Rd

Op Code Arguments





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***who** could be interested in this?*

anybody who ...

... wants to know why and how computer science
immediately connects and translates to the physical world.

... would like to see **immediate real-world involvement** in their work.

... would like to understand what
really happens if you run a high level program.



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who are these people? – introductions

Charles Martin & Uwe R. Zimmer

Aditya (**Adi**) Chilukuri, Benjamin (**Ben**) Cartwright,

Brent (**Brentles**) Schuetze, **Calum** Snowdon,

Chinmay Garg, **Cody** Christopher, Joshua (**Josh**) Corner,

Joshua (**Josh**) Gilbert, Mitchell (**Mitch**) Rowett, **Septian** Razi,

Timothy (**Tim**) Lee,

William (**Will**) Cashman,

Yutong (**Alice**) Wang,

Zhuo (**Zoey**) Chen



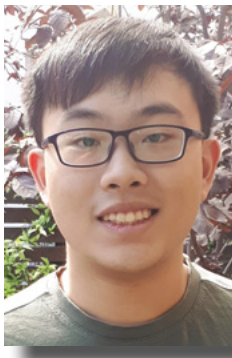
Charles



Uwe



Alice



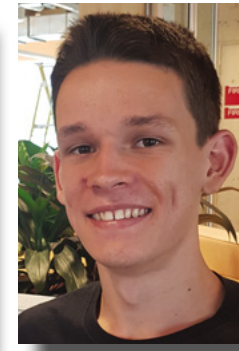
Tim



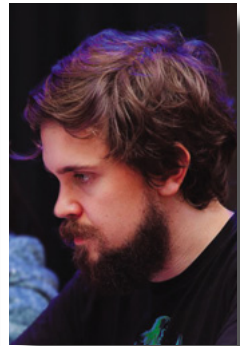
Josh G.



Cody



Will



Brentles



Adi



Josh C.



Calum



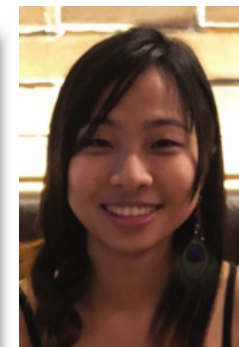
Ben



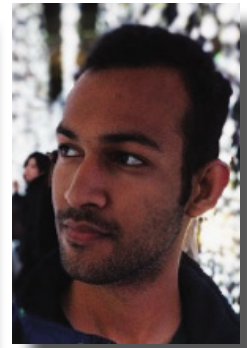
Mitch



Septian



Zoey



Chinmay



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***how** will this all be done?*

☞ Lectures:

- 2x 1.5 hours lectures per week ... all the nice stuff
Wednesday 15:00, Thursday 10:00 (both in Coombs Theatre - which is: *here*)

☞ Laboratories:

- 3 hours per week ... all the rough stuff
time slots: on our web-site – all in CSIT N.xxx or HN Lab.xx laboratories
-enrolment: <https://cs.anu.edu.au/streams/> (opened on Monday)

☞ Resources:

- Course site: <http://cs.anu.edu.au/student/comp2300/> ... as well as schedules, slides, sources, links to forums, etc. pp. ... keep an eye on this page!

☞ Assessment:

- *Hurdle lab* in week 4 (1%) – a pass here is a **hurdle** for the course
- *Mid-semester exam* (13%)
- *3 assignments* (12% each)
- *Final-exam* at the end of the course (50%) – 40/100 is a **hurdle** for the final exam



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“Text book” for the course

[Patterson17]

David A. Patterson & John L. Hennessy

Computer Organization and Design – The Hardware/Software Interface

ARM edition, Morgan Kaufmann 2017

☞ Many concepts in this course are in there – *but not all!*

The [Patterson17] provides an excellent general background and a lot of in-depth studies into more specific fields.

☞ *References for specific aspects of the course are provided during the course and are found on our web-site.*